

Week_5_lab

CHAKAR

2026-07-31

Week 5 lab - Deliverable 3

This file reads the monthly Airbnb listings CSVs, filters them to Christchurch City, combines them into one dataset, and explores price and review patterns.

Task: Filter on Christchurch data only. Use only this filtered dataset from now on. Using 'june_2026_listings.csv' first

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
## Rows: 50932 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr   (5): name, host_name, neighbourhood_group, neighbourhood, room_type
## dbl   (11): id, host_id, latitude, longitude, price, minimum_nights, number_o...
## lgl   (1): license
## date  (1): last_review
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

## [1] "The first 10 rows are:"

## # A tibble: 10 x 18
##       id name      host_id host_name neighbourhood_group neighbourhood latitude
##   <dbl> <chr>      <dbl> <chr>      <chr>              <chr>         <dbl>
## 1 1.52e18 Back-ya~ 7.22e8 David    Waikato District  Ngauwawahia ~ -37.7
## 2 7.87e17 Nice n ~ 2.28e8 Shallu  Hamilton City    East Ward        -37.7
## 3 1.20e7  Manarvi~ 3.27e7          Waikato District  Huntly Ward      -37.6
## 4 7.64e17 Room in~ 1.62e8 Simranjot Matamata-Piako Dis~ Morrinsville~ -37.6
## 5 1.29e18 Brand N~ 3.72e8 Matt    Hamilton City    East Ward        -37.7
## 6 1.60e18 Flat 8C  5.63e8 Godfrey  South Waikato Dist~ Tokoroa Ward    -38.2
## 7 1.62e18 Family ~ 7.19e8 Pukawa ~  Taupo District  Turangi-Tong~ -38.9
```

```
## 8 1.62e18 Family ~ 7.19e8 Pukawa ~ Taupo District Turangi-Tong~ -38.9
## 9 1.49e18 Apartme~ 4.69e7 Craig Ashburton District Ashburton Wa~ -43.9
## 10 1.11e18 The Cos~ 5.66e8 Melissa Stratford District Stratford Ur~ -39.3
## # i 11 more variables: longitude <dbl>, room_type <chr>, price <dbl>,
## # minimum_nights <dbl>, number_of_reviews <dbl>, last_review <date>,
## # reviews_per_month <dbl>, calculated_host_listings_count <dbl>,
## # availability_365 <dbl>, number_of_reviews_ltm <dbl>, license <lgl>
```

```
## [1] "Summary of the dataset is:"
```

```
##      id                name      host_id      host_name
## Min.   :3.619e+04   Length   :50932   Min.    :2.930e+04   Length   :50932
## 1st Qu.:4.514e+07   N.unique :49673   1st Qu.:6.001e+07   N.unique : 8649
## Median :1.012e+18   N.blank  :    0   Median :1.523e+08   N.blank  :    0
## Mean   :8.359e+17   Min.nchar:    1   Mean   :6.582e+15   Min.nchar:    1
## 3rd Qu.:1.397e+18   Max.nchar:  138   3rd Qu.:3.749e+08   Max.nchar:   47
## Max.    :1.711e+18                                     Max.    :1.711e+18   NAs      :    6
##                                     NAs      :1
## neighbourhood_group neighbourhood      latitude      longitude
## Length   :50932     Length   :50932   Min.    :-46.91   Min.    :-176.6
## N.unique :    67     N.unique :   210   1st Qu.: -43.54   1st Qu.:  172.6
## N.blank  :    0     N.blank  :    0   Median : -39.34   Median :  174.7
## Min.nchar:    8     Min.nchar:    9   Mean    :-40.19   Mean    :  173.5
## Max.nchar:   30     Max.nchar:   48   3rd Qu.: -36.94   3rd Qu.:  175.4
##                                     Max.    :-34.52   Max.    :  178.5
##
##      room_type      price      minimum_nights      number_of_reviews
## Length   :50932   Min.    :    5.0   Min.    :  1.000   Min.    :  0.00
## N.unique :    4   1st Qu.:  173.0   1st Qu.:  1.000   1st Qu.:  4.00
## N.blank  :    0   Median :  264.0   Median :  1.000   Median :  22.00
## Min.nchar:   10   Mean    :  378.2   Mean    :  2.104   Mean    :  67.33
## Max.nchar:   15   3rd Qu.:  429.0   3rd Qu.:  2.000   3rd Qu.:  78.00
##                                     Max.    :84438.0   Max.    :639.000   Max.    :1584.00
##                                     NAs      :5223     NAs      :8
## last_review      reviews_per_month      calculated_host_listings_count
## Min.    :2013-03-03   Min.    : 0.010   Min.    :  1.0
## 1st Qu.:2026-02-11   1st Qu.: 0.420   1st Qu.:  1.0
## Median :2026-04-18   Median : 1.110   Median :  2.0
## Mean    :2026-01-16   Mean    : 1.750   Mean    : 41.6
## 3rd Qu.:2026-05-31   3rd Qu.: 2.413   3rd Qu.:  6.0
## Max.    :2026-06-22   Max.    :24.720   Max.    :1215.0
## NAs     :5128        NAs     :5128     NAs     :1
## availability_365      number_of_reviews_ltm      license
## Min.    : 0.0        Min.    : 0.00        Mode:logical
## 1st Qu.: 93.0        1st Qu.: 1.00        NAs :50932
## Median :227.0        Median : 7.00
## Mean    :206.5        Mean    :14.83
## 3rd Qu.:320.0        3rd Qu.:20.00
## Max.    :365.0        Max.    :267.00
##
```

```
## [1] "Waikato District"      "Hamilton City"
## [3] "Matamata-Piako District" "South Waikato District"
```

## [5] "Taupo District"	"Ashburton District"
## [7] "Stratford District"	"Whangarei District"
## [9] "Marlborough District"	"New Plymouth District"
## [11] "Auckland"	"Christchurch City"
## [13] "Dunedin City"	"Invercargill City"
## [15] "Ruapehu District"	"Wairoa District"
## [17] "Waimakariri District"	"Waitomo District"
## [19] "Tasman District"	"Kaipara District"
## [21] "Horowhenua District"	"Selwyn District"
## [23] "South Wairarapa District"	"Napier City"
## [25] "Palmerston North City"	"Whakatane District"
## [27] "Queenstown-Lakes District"	"Central Otago District"
## [29] "Southland District"	"Rangitikei District"
## [31] "Far North District"	"Thames-Coromandel District"
## [33] "Kaikoura District"	"Mackenzie District"
## [35] "Ōtorohanga District"	"Carterton District"
## [37] "Tauranga City"	"Hauraki District"
## [39] "Western Bay of Plenty District"	"Waipa District"
## [41] "Gore District"	"Westland District"
## [43] "Rotorua District"	"Clutha District"
## [45] "Lower Hutt City"	"Masterton District"
## [47] "Central Hawke's Bay District"	"Hurunui District"
## [49] "Waitaki District"	"Hastings District"
## [51] "Nelson City"	"Timaru District"
## [53] "Buller District"	"Kapiti Coast District"
## [55] "Gisborne District"	"Wellington City"
## [57] "Porirua City"	"Waimate District"
## [59] "Ōpōtiki District"	"Whanganui District"
## [61] "Manawatu District"	"South Taranaki District"
## [63] "Grey District"	"Taranua District"
## [65] "Kawerau District"	"Upper Hutt City"
## [67] "Chatham Islands Territory"	
## [1] "Waikato District"	"Hamilton City"
## [3] "Matamata-Piako District"	"South Waikato District"
## [5] "Taupo District"	"Ashburton District"
## [7] "Stratford District"	"Whangarei District"
## [9] "Marlborough District"	"New Plymouth District"
## [11] "Auckland"	"Christchurch City"
## [13] "Dunedin City"	"Invercargill City"
## [15] "Ruapehu District"	"Wairoa District"
## [17] "Waimakariri District"	"Waitomo District"
## [19] "Tasman District"	"Kaipara District"
## [21] "Horowhenua District"	"Selwyn District"
## [23] "South Wairarapa District"	"Napier City"
## [25] "Palmerston North City"	"Whakatane District"
## [27] "Queenstown-Lakes District"	"Central Otago District"
## [29] "Southland District"	"Rangitikei District"
## [31] "Far North District"	"Thames-Coromandel District"
## [33] "Kaikoura District"	"Mackenzie District"
## [35] "Ōtorohanga District"	"Carterton District"
## [37] "Tauranga City"	"Hauraki District"
## [39] "Western Bay of Plenty District"	"Waipa District"
## [41] "Gore District"	"Westland District"

```
## [43] "Rotorua District"           "Clutha District"
## [45] "Lower Hutt City"            "Masterton District"
## [47] "Central Hawke's Bay District" "Hurunui District"
## [49] "Waitaki District"          "Hastings District"
## [51] "Nelson City"                "Timaru District"
## [53] "Buller District"            "Kapiti Coast District"
## [55] "Gisborne District"          "Wellington City"
## [57] "Porirua City"               "Waimate District"
## [59] "Ōpōtiki District"           "Whanganui District"
## [61] "Manawatu District"          "South Taranaki District"
## [63] "Grey District"              "Taranaki District"
## [65] "Kawerau District"           "Upper Hutt City"
## [67] "Chatham Islands Territory"
```

Checking process by filtering june before full chch data

Filtering all the data sets to Christchurch city and adding month + year column using a pipeline

Applying the load_chch_listings function to each month's dataset, filtering to Christchurch City and tagging each with its month_year.

```
## Rows: 47752 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr   (5): name, host_name, neighbourhood_group, neighbourhood, room_type
## dbl  (11): id, host_id, latitude, longitude, price, minimum_nights, number_o...
## lgl   (1): license
## date  (1): last_review
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 48624 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr   (5): name, host_name, neighbourhood_group, neighbourhood, room_type
## dbl  (11): id, host_id, latitude, longitude, price, minimum_nights, number_o...
## lgl   (1): license
## date  (1): last_review
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 49899 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr   (5): name, host_name, neighbourhood_group, neighbourhood, room_type
## dbl  (10): id, host_id, latitude, longitude, minimum_nights, number_of_revie...
## lgl   (2): price, license
## date  (1): last_review
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 46819 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr   (5): name, host_name, neighbourhood_group, neighbourhood, room_type
```

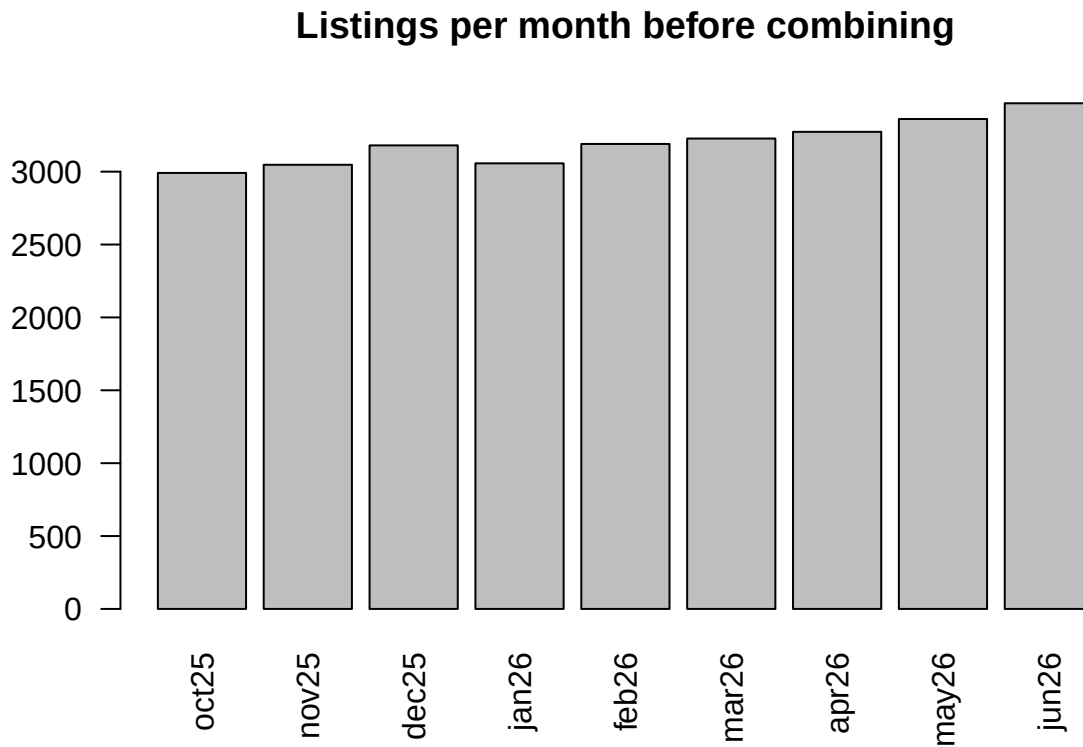
```

## dbl (10): id, host_id, latitude, longitude, minimum_nights, number_of_revie...
## lgl (2): price, license
## date (1): last_review
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 48692 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr (5): name, host_name, neighbourhood_group, neighbourhood, room_type
## dbl (10): id, host_id, latitude, longitude, minimum_nights, number_of_revie...
## lgl (2): price, license
## date (1): last_review
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 49076 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr (5): name, host_name, neighbourhood_group, neighbourhood, room_type
## dbl (11): id, host_id, latitude, longitude, price, minimum_nights, number_o...
## lgl (1): license
## date (1): last_review
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 48951 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr (5): name, host_name, neighbourhood_group, neighbourhood, room_type
## dbl (11): id, host_id, latitude, longitude, price, minimum_nights, number_o...
## lgl (1): license
## date (1): last_review
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 49305 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr (5): name, host_name, neighbourhood_group, neighbourhood, room_type
## dbl (11): id, host_id, latitude, longitude, price, minimum_nights, number_o...
## lgl (1): license
## date (1): last_review
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 50932 Columns: 18
## -- Column specification -----
## Delimiter: ","
## chr (5): name, host_name, neighbourhood_group, neighbourhood, room_type
## dbl (11): id, host_id, latitude, longitude, price, minimum_nights, number_o...
## lgl (1): license
## date (1): last_review
##

```

```
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Sanity check: bar chart of row counts per month, check if there are missing/extra data across months - check all 9 months are loaded and filtered successfully, with a similar number of chch listings each month



Combining all nine monthly datasets into a single Christchurch dataset spanning October 2025 to June 2026.

Summary stats

```
## # A tibble: 7 x 5
##   variable      min      max    mean    sd
##   <chr>      <dbl>    <dbl>  <dbl>  <dbl>
## 1 price      27    43654   260.   944.
## 2 minimum_nights      1     365    1.79   5.51
## 3 number_of_reviews      0    1584    70.4   121.
## 4 reviews_per_month    0.01   16.6    2.35   2.19
## 5 calculated_host_listings_count      1    1586    58.3   189.
## 6 availability_365      0     365   204.   112.
## 7 number_of_reviews_ltm      0     185    19.0   23.5

## [1] 28795

##
## Entire home/apt      Hotel room      Private room      Shared room
##           22765              251             5744              35
```

```
##
## Banks Peninsula Ward      Burwood Ward      Cashmere Ward
##           3548           372           1009
##           Central Ward    Coastal Ward      Fendalton Ward
##           10554           891           1605
##           Halswell Ward   Harewood Ward  Heathcote Ward
##           1081           909           2074
##           Hornby Ward     Innes Ward     Linwood Ward
##           594            1027           659
##           Papanui Ward    Riccarton Ward  Spreydon Ward
##           648            1885           1056
##           Waimairi Ward
##           883
```

```
##           id           name
##           0           0
##           host_id      host_name
##           0           1
##           neighbourhood_group  neighbourhood
##           0           0
##           latitude      longitude
##           0           0
##           room_type     price
##           0           10667
##           minimum_nights  number_of_reviews
##           37           0
##           last_review    reviews_per_month
##           2627          2627
## calculated_host_listings_count  availability_365
##           0           0
##           number_of_reviews_ltm  license
##           0           28795
##           month_year
##           0
```

Storing the combined Christchurch dataset named: all_chch_listings.csv

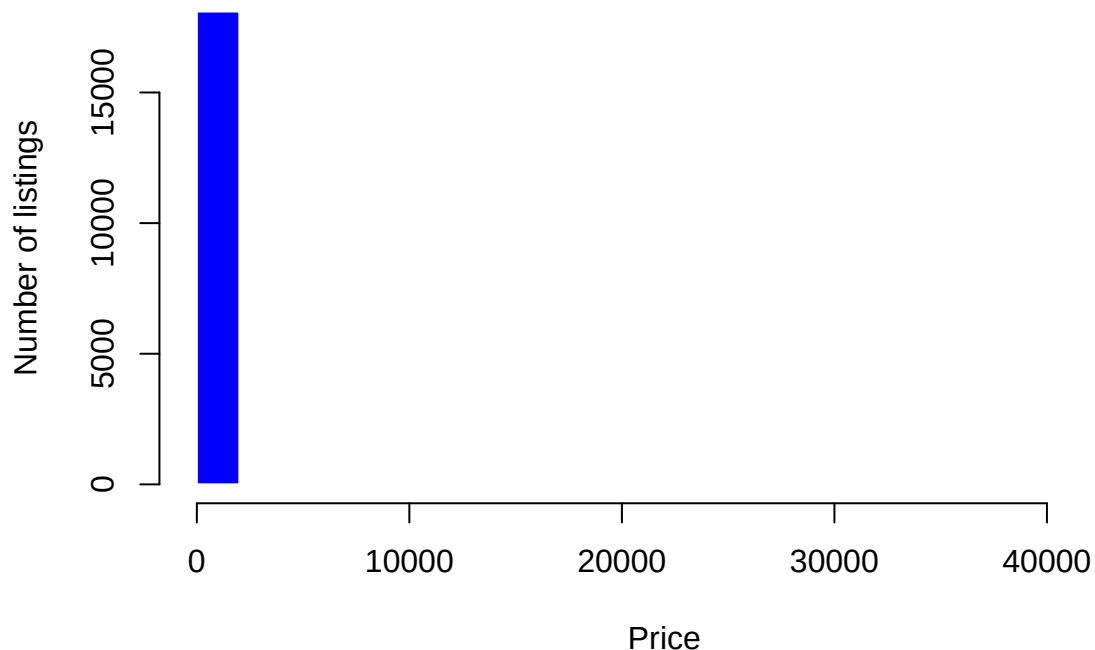
```
## Rows: 28795 Columns: 19
## -- Column specification -----
## Delimiter: ","
## chr  (6): name, host_name, neighbourhood_group, neighbourhood, room_type, m...
## dbl  (11): id, host_id, latitude, longitude, price, minimum_nights, number_o...
## lgl  (1): license
## date (1): last_review
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

## # A tibble: 10 x 19
##       id name      host_id host_name neighbourhood_group neighbourhood latitude
##   <dbl> <chr>    <dbl> <chr>    <chr>                <chr>         <dbl>
## 1 4.02e 7 Ensuite~ 9.88e7 North So~ Christchurch City  Harewood Ward -43.5
## 2 1.15e18 Handy c~ 2.98e8 Tony      Christchurch City  Central Ward   -43.5
```

```
## 3 1.10e 6 Heritag~ 4.99e6 Paulette Christchurch City Fendalton Wa~ -43.5
## 4 1.34e18 Settle ~ 5.43e7 Sarena Christchurch City Burwood Ward -43.5
## 5 1.21e18 Eco Spa~ 1.79e8 Peng Christchurch City Riccarton Wa~ -43.5
## 6 1.45e18 Merival~ 1.54e8 Alisa Ka~ Christchurch City Fendalton Wa~ -43.5
## 7 5.27e 7 Studio ~ 4.26e8 Raj Christchurch City Riccarton Wa~ -43.5
## 8 8.10e17 City li~ 4.97e8 Michael Christchurch City Heathcote Wa~ -43.6
## 9 3.94e 7 Wigram ~ 3.03e8 Hongmei Christchurch City Halswell Ward -43.6
## 10 1.46e18 Dollar ~ 2.98e8 Tony Christchurch City Heathcote Wa~ -43.6
## # i 12 more variables: longitude <dbl>, room_type <chr>, price <dbl>,
## # minimum_nights <dbl>, number_of_reviews <dbl>, last_review <date>,
## # reviews_per_month <dbl>, calculated_host_listings_count <dbl>,
## # availability_365 <dbl>, number_of_reviews_ltm <dbl>, license <lgl>,
## # month_year <chr>
```

Task Reproduce the workflow and the plots in R Plot the distribution of price as a histogram for christchurch city

Distribution of Price for Christchurch City (with outliers)



```
## price_bins
## (0,5e+03] (5e+03,1e+04] (1e+04,1.5e+04] (1.5e+04,2e+04] (2e+04,2.5e+04]
## 18115 4 0 0 1
## (2.5e+04,3e+04] (3e+04,3.5e+04] (3.5e+04,4e+04] (4e+04,4.5e+04]
## 0 0 0 8
```

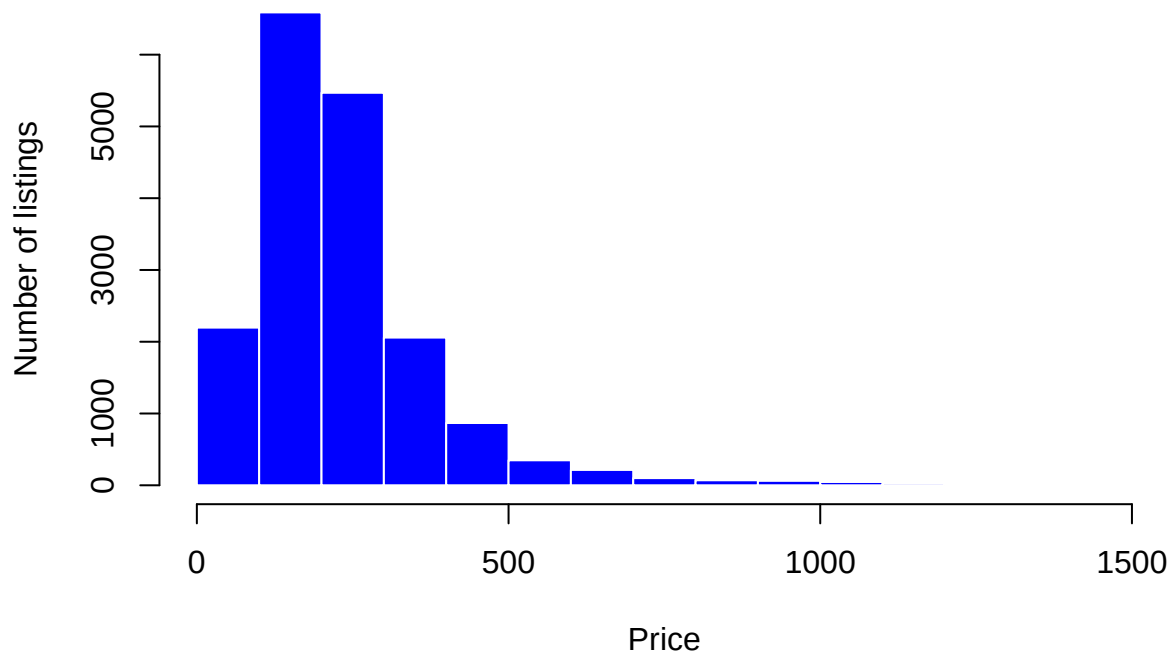
1 listing priced between 20000-25000, and 8 priced at 40000-45000

```
## # A tibble: 6 x 19
```

```
##      id name      host_id host_name neighbourhood_group neighbourhood latitude
##      <dbl> <chr>      <dbl> <chr>      <chr>      <chr>      <dbl>
## 1 5.94e17 Seaview ~ 1.45e8 Bachcare Christchurch City Banks Penins~ -43.8
## 2 1.41e18 Central ~ 1.45e8 Bachcare Christchurch City Central Ward -43.5
## 3 9.15e17 La Maiso~ 1.45e8 Bachcare Christchurch City Banks Penins~ -43.8
## 4 1.00e18 Pacific ~ 1.45e8 Bachcare Christchurch City Central Ward -43.5
## 5 9.15e17 La Maiso~ 1.45e8 Bachcare Christchurch City Banks Penins~ -43.8
## 6 5.94e17 Seaview ~ 1.45e8 Bachcare Christchurch City Banks Penins~ -43.8
## # i 12 more variables: longitude <dbl>, room_type <chr>, price <dbl>,
## #   minimum_nights <dbl>, number_of_reviews <dbl>, last_review <date>,
## #   reviews_per_month <dbl>, calculated_host_listings_count <dbl>,
## #   availability_365 <dbl>, number_of_reviews_ltm <dbl>, license <lgl>,
## #   month_year <chr>
```

To check the accommodation priced at \$22971/night. Checking its latitude (-43.52059) and longitude(172.6496) showed on google map to be at address 333 Bealey Avenue, chch central. Checked accommodation on google map priced at \$43188/night and \$43178/night, also appeared unlikely to worth that much. These 9 values likely error, to drop these 9 extreme outlier values

Distribution of Price for Christchurch city (without outliers)



Task- Calculate how long ago the last review was in days (difference between scrape/publish date and last review date). Plot the distribution of this new feature.

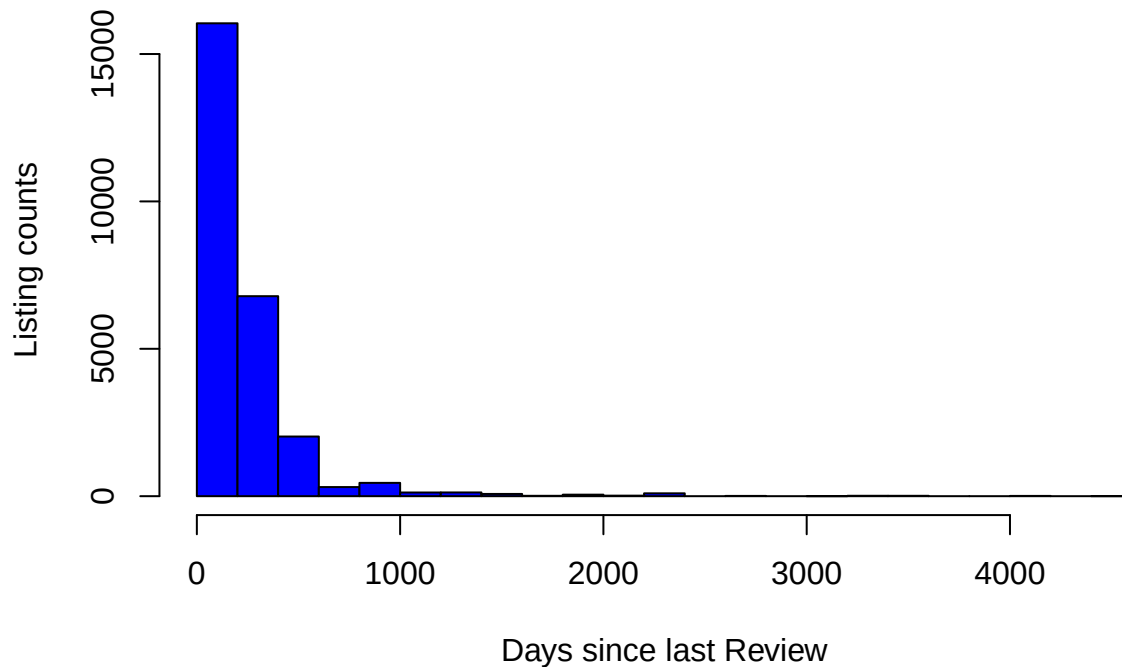
(go back to use the full chch dataset variable name: chch_file, as reviews may still be correct)

```
## [1] "Date"

## [1] NA "2025-07-15" "2025-03-03" "2025-04-26" "2025-03-01"
```

```
## [6] NA
```

Distribution of Days since Last Review



Task - Find out which properties in New Zealand get the highest numbers of reviews (filter top 10%). How many of them are in Christchurch?

Filtering the data by decreasing

```
## [1] 1584 1570 1555 1551 1544 1542 1528 1525 1518 1516 1509 1498 1495 1489 1484
## [16] 1482 1474 1465 1393 1389 1385 1380 1369 1359 1349 1337 1325 1160 1150 1142
## [31] 1139 1122 1110 1095 1095 1095 1077 1073 1064 1050
```

```
## # A tibble: 28,795 x 20
```

```
##       id name      host_id host_name neighbourhood_group neighbourhood latitude
##       <dbl> <chr>      <dbl> <chr>      <chr>              <chr>          <dbl>
## 1  6467168 lazy p~  3.38e7 Lou      Christchurch City  Linwood Ward   -43.5
## 2  6467168 lazy p~  3.38e7 Lou      Christchurch City  Linwood Ward   -43.5
## 3  6467168 lazy p~  3.38e7 Lou      Christchurch City  Linwood Ward   -43.5
## 4 12613591 Airpor~  6.75e7 Sharon    Christchurch City  Papanui Ward   -43.5
## 5 12613591 Airpor~  6.75e7 Sharon    Christchurch City  Papanui Ward   -43.5
## 6  6467168 lazy p~  3.38e7 Lou      Christchurch City  Linwood Ward   -43.5
## 7 12613591 Airpor~  6.75e7 Sharon    Christchurch City  Papanui Ward   -43.5
## 8  6467168 lazy p~  3.38e7 Lou      Christchurch City  Linwood Ward   -43.5
## 9 12613591 Airpor~  6.75e7 Sharon    Christchurch City  Papanui Ward   -43.5
## 10 6467168 lazy p~  3.38e7 Lou      Christchurch City  Linwood Ward   -43.5
## # i 28,785 more rows
## # i 13 more variables: longitude <dbl>, room_type <chr>, price <dbl>,
```

```
## # minimum_nights <dbl>, number_of_reviews <dbl>, last_review <date>,
## # reviews_per_month <dbl>, calculated_host_listings_count <dbl>,
## # availability_365 <dbl>, number_of_reviews_ltm <dbl>, license <lgl>,
## # month_year <chr>, difference_days <dbl>
```

Finding the number of rows in the dataset and finding 10% (0.1) of the data

```
## [1] 2879.5
```

Showing top 10% of the data

```
## # A tibble: 2,879 x 20
##       id name      host_id host_name neighbourhood_group neighbourhood latitude
##       <dbl> <chr>      <dbl> <chr>      <chr>      <chr>      <dbl>
## 1  6467168 lazy p~  3.38e7 Lou      Christchurch City Linwood Ward -43.5
## 2  6467168 lazy p~  3.38e7 Lou      Christchurch City Linwood Ward -43.5
## 3  6467168 lazy p~  3.38e7 Lou      Christchurch City Linwood Ward -43.5
## 4 12613591 Airpor~  6.75e7 Sharon    Christchurch City Papanui Ward -43.5
## 5 12613591 Airpor~  6.75e7 Sharon    Christchurch City Papanui Ward -43.5
## 6  6467168 lazy p~  3.38e7 Lou      Christchurch City Linwood Ward -43.5
## 7 12613591 Airpor~  6.75e7 Sharon    Christchurch City Papanui Ward -43.5
## 8  6467168 lazy p~  3.38e7 Lou      Christchurch City Linwood Ward -43.5
## 9 12613591 Airpor~  6.75e7 Sharon    Christchurch City Papanui Ward -43.5
## 10 6467168 lazy p~  3.38e7 Lou      Christchurch City Linwood Ward -43.5
## # i 2,869 more rows
## # i 13 more variables: longitude <dbl>, room_type <chr>, price <dbl>,
## # minimum_nights <dbl>, number_of_reviews <dbl>, last_review <date>,
## # reviews_per_month <dbl>, calculated_host_listings_count <dbl>,
## # availability_365 <dbl>, number_of_reviews_ltm <dbl>, license <lgl>,
## # month_year <chr>, difference_days <dbl>
```